$G(6, \mathbb{Z}_3\mathbb{C})$ Vacuum Structure and FFGFT

Numerical Investigation of the Connection Questions

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Contents

List of Symbols

G(6,$\mathbb{Z}_3\mathbb{C}$) Symbols after Matzke (2026)

Symbol	Meaning	Remark
$G(6,\mathbb{Z}_3\mathbb{C})$	Clifford algebra, 6 dimensions, coefficient field $\{-1,0,\pm1,\pm i\}$	$2 \equiv -1 \pmod 3$
$e_1,\dots,e_6$	Basis vectors, $e_k^2 = +1$	
$N_k^\pm$	Nilpotent vectors, $(N_k^\pm)^2 = 0$	$N_k^\pm = e_a \pm i\, e_b$
P_{Pk}	Idempotents, $P_{Pk}^2 = P_{Pk}$	$P_{Pk} = (\mathbf{1} + i\, e_a e_b)/2$
V	Vacuum operator, $V = P_{P1}\cdot P_{P2}\cdot P_{P3}$	rank-1 projector
T_k	Trine operator, $T_k^3 = \mathbf{1}$	$T_k = \mathbf{1} + (\omega-1)P_{Pk}$
ω	Primitive $\mathbb{Z}_3$ phase factor	$e^{2\pi i/3}$
I,J,K	Commutative quaternions (grade 4), $I^2 = J^2 = K^2 = +1$	encode 3 spatial dimensions
iE_7	Complex volume form, $(iE_7)^2 = +1$	universal phase operator

FFGFT Symbols after Pascher (2023–2026)

Symbol	Meaning	Value
ξ	FFGFT geometry parameter	$4/30000$
$T^4/\mathbb{Z}_3$	4D torus with $\mathbb{Z}_3$ orbifold projection	
τ	Generator of the $\mathbb{Z}_3$ action on T^4	unitary
P_k	$\mathbb{Z}_3$ projectors on $L^2(T^4)$	$\frac{1}{3}\sum_j \omega^{-jk}\tau^j$, $k = 0, 1, 2$
$\mathscr{H}_k$	Eigensectors: $\mathscr{H}_k = P_k L^2(T^4)$	three colour sectors
$\hat{F}_{D_4}$	D_4 sub-operator of the spectral operator	$\xi = \lambda_{\text{min}}(\hat{F}_{D_4})$
$C_2(R)$	Quadratic Casimir operator (Lie algebra)	$C_2(SU(3)_{\text{fund}}) = 4/3$
N_{Fourier}	Fourier mode count of the T^4 torus	$10^4 = 30000/\lvert\mathbb{Z}_3\rvert$
N_c	Colour degrees of freedom, algebraically necessary	3
v	Higgs vacuum expectation value	246 GeV

FFGFT Document Numbering

This document references the FFGFT corpus (Johann Pascher, 2023–2026, ORCID 0009-0000-6518-4064):

- **Doc. XXX**: Document No. XXX of the FFGFT corpus. Full list: `github.com/jpascher/T0-Time-Mass-Duality`
- **RXXX**: Correction/addendum No. XXX in the register Doc. 190.
- **[B]**: algebraically proved **[K]**: derivable from ξ **[S]**: sketched/plausible

Documents relevant to this comparison: Doc. 009 (Casimir effect, factor $4/3$), Doc. 049 (confinement), Doc. 160 (α_s), Doc. 314 (D_4 lattice), Doc. 321 ($SU(3)_c$ derivation), Doc. 322 (Hilbert space), Doc. 323 (Weinberg angle).

Summary

This document compares the vacuum structure of $G(6, \mathbb{Z}_3\mathbb{C})$ after Matzke [1] with FFGFT spectral theory and numerically investigates the open connection question: is $\xi = 4/30000$ the smallest spectral value of the vacuum operator V in G(6)?

Results of the numerical investigation (Python script `324_G6_FFGFT_verify.py`):

- The Trine theorem is verified: $T_k^3 = \mathbf{1}$ to 6.5×10^{-16} **[B]**.
- The vacuum operator $V = P_{P1} \cdot P_{P2} \cdot P_{P3}$ is a rank-1 projector: $V^2 = V$ with eigenvalues $\{0^{(7)},\ 1^{(1)}\}$.
- ξ is *not* a direct spectral value of V. The connection is *structural*, not spectral.
- The trine product $T_1T_2T_3$ is a global $\mathbb{Z}_3$ operator: $(T_1T_2T_3)^3 = \mathbf{1}$, eigenvalues $\{1^{(2)}, \omega^{(3)}, \omega^{2(3)}\}$ **[B]**.
- $\xi = C_2(SU(3)_{\text{fund}})/N_{\text{Fourier}} = (4/3)/10^4$ algebraically derived – R84 **[K]**.

Status markers

[K] derivable from ξ **[B]** algebraically proved **[S]** sketched/plausible

.1 Starting Point: Two Frameworks, One Physics

Matzke's G(6,$\mathbb{Z}_3\mathbb{C}$) Vacuum Structure

[S] Matzke constructs the physical vacuum state V as an algebraic object in $G(6, \mathbb{Z}_3\mathbb{C})$, a Clifford algebra over the coefficient field $\mathbb{Z}_3\mathbb{C} = \{-1, 0, +1, \pm i\}$. The central vacuum relation is

$$V^2 = -V \quad (\text{in } \mathbb{Z}_3\mathbb{C} \text{ arithmetic}), \tag{1}$$

where $2 \equiv -1 \pmod 3$. Three nilpotent vector pairs $N_k^\pm = e_a \pm i\, e_b$ with $(N_k^\pm)^2 = 0$ decompose the $\mathbb{C}^6$ spinor space into three eigensectors. The Trine theorem states: for every nilpotent N with $N^2 = 0$,

$$T = \mathbf{1} + (\omega - 1)N \quad \Rightarrow \quad T^3 = \mathbf{1}. \tag{2}$$

FFGFT Spectral Theory

[K] In FFGFT, $\xi = 4/30000$ is the smallest eigenvalue of the D_4 sub-operator $\hat{F}_{D_4}$ (Doc. 322):

$$\xi = \lambda_{\text{min}}(\hat{F}_{D_4}) = \frac{4}{30000}. \tag{3}$$

The $\mathbb{Z}_3$ projection $P_k = \frac{1}{3}\sum_{j=0}^{2} \omega^{-jk}\tau^j$ on $L^2(T^4)$ decomposes the Hilbert space into three eigensectors $\mathscr{H}_0, \mathscr{H}_1, \mathscr{H}_2$. Their bilinear transitions generate the $\mathfrak{su}(3)$ algebra (Doc. 321, **[B]**).

.2 Numerical Verification of the Trine Theorem

Construction of the Vacuum Structure

We use the 8-dimensional spinor representation of G(6) over $\mathbb{C}$, given by Γ matrices as tensor products of Pauli matrices:

$$\begin{aligned} &\Gamma_1 = \sigma_x \otimes \mathbf{1} \otimes \mathbf{1}, \quad \Gamma_2 = \sigma_y \otimes \mathbf{1} \otimes \mathbf{1}, \quad \Gamma_3 = \sigma_z \otimes \sigma_x \otimes \mathbf{1}, \\ &\Gamma_4 = \sigma_z \otimes \sigma_y \otimes \mathbf{1}, \quad \Gamma_5 = \sigma_z \otimes \sigma_z \otimes \sigma_x, \quad \Gamma_6 = \sigma_z \otimes \sigma_z \otimes \sigma_y. \end{aligned} \tag{4}$$

The Clifford relation $\{\Gamma_i, \Gamma_j\} = 2\delta_{ij}\mathbf{1}_8$ holds (numerically to $< 10^{-14}$).

Nilpotents over three Witt pairs $(0,1)$, $(2,3)$, $(4,5)$:

$$N_k^+ = \Gamma_{2k} + i\,\Gamma_{2k+1}, \quad N_k^- = \Gamma_{2k} - i\,\Gamma_{2k+1}, \quad (N_k^\pm)^2 = 0 \quad \textbf{[B]}. \tag{5}$$

Idempotents:

$$P_{Pk} = \frac{\mathbf{1} + i\,\Gamma_{2k}\Gamma_{2k+1}}{2}, \qquad P_{Pk}^2 = P_{Pk} \quad (\text{numerically to } 0) \quad \textbf{[B]}. \tag{6}$$

Trine Operators and Their Spectrum

With $\omega = e^{2\pi i/3}$:

$$T_k = \mathbf{1} + (\omega - 1)\, P_{Pk}. \tag{7}$$

[B] Numerical result:

$$T_k^3 = \mathbf{1} \quad \text{to } 6.5 \times 10^{-16}. \tag{8}$$

The Trine theorem is exactly verified in the 8-dimensional spinor representation of G(6).

The trine product:

$$(T_1 T_2 T_3)^3 = \mathbf{1} \quad \text{to } 1.9 \times 10^{-15}. \tag{9}$$

Eigenvalues of $T_1T_2T_3$: $\{1^{(2)}, \omega^{(3)}, \omega^{2(3)}\}$ – a global $\mathbb{Z}_3$ phase operator on the 8-dimensional spinor space.

.3 Spectrum of the Vacuum Operator V

[B] The vacuum operator $V = P_{P1} \cdot P_{P2} \cdot P_{P3}$ has spectrum:

$$\sigma(V) = \{0^{(7)},\ 1^{(1)}\}. \tag{10}$$

V is a rank-1 projector. In $\mathbb{Z}_3\mathbb{C}$ arithmetic, $V^2 = V$ with $V^2 \equiv -V \pmod 3$, reproducing Matzke's relation $V^2 = -V$.

[K] $\xi = 4/30000$ is *not* a spectral value of V. The connection is structural:

Table 1: Structural comparison $G(6, \mathbb{Z}_3\mathbb{C})$ vs. FFGFT $T^4/\mathbb{Z}_3$

Quantity	$G(6, \mathbb{Z}_3\mathbb{C})$	FFGFT $T^4/\mathbb{Z}_3$
$\mathbb{Z}_3$ symmetry	Witt decomposition of $\mathbb{C}^6$	Orbifold projection on T^4
Three sectors	three nilpotent pairs $N_k^{\pm}$	$\mathscr{H}_0, \mathscr{H}_1, \mathscr{H}_2$
$N_c = 3$	number of Witt pairs	order of $\mathbb{Z}_3$
Confinement	$T_R = 0$ (triality)	$T_R = 0$, algebraic [B]
Vacuum	rank-1 projector V	spectral value $\xi = \lambda_{\text{min}}(\hat{F}_{D_4})$
$SU(3)_c$	bilinear transitions [S]	proved [B] (Doc. 321)
Fermion masses	nilpotents as directions	eigenvalues $m_i = r_i \xi^{p_i} v$ [K]
Numerator 4 in ξ	4 Witt pairs in $G(8)$	4 torus dimensions
Denominator 30000	$\lvert\mathbb{Z}_3\rvert \times 10^4$ (modes)	$\lvert\mathbb{Z}_3\rvert \times 10^4$ (modes)

.4 Agreements and Divergences

[B] The following structures are identical in both frameworks and numerically verified:

1. **Trine theorem:** $T_k^3 = \mathbf{1}$ from nilpotents $N_k^{\pm}$ – identical with the $\mathbb{Z}_3$ phase structure of FFGFT projectors $P_k = \frac{1}{3} \sum \omega^{-jk} \tau^j$.
2. **Three sectors from one symmetry:** G(6) Witt decomposition ↔ FFGFT $\mathbb{Z}_3$ eigenspace decomposition. Both yield exactly three sectors; their bilinear transitions generate $\mathfrak{su}(3)$.
3. **Confinement as $T_R = 0$:** Matzke's complement condition $V(V + 1) = 0$ corresponds to the triality selection in Doc. 321.
4. **$N_c = 3$ algebraically forced:** Neither G(6) nor FFGFT sets N_c as a free parameter.

Table 2: Open and closed bridges between $G(6, \mathbb{Z}_3\mathbb{C})$ and FFGFT

Point	$G(6, \mathbb{Z}_3\mathbb{C})$	FFGFT	Character
Electroweak group	$SU(2)_L$ open [S]	$SU(2)_L$, $U(1)_Y$ [B]	G(6)-internal
Weinberg angle	not computed	0.2308 ($-0.19\,\%$) [K]	G(6)-internal
Lepton masses	not derived	$m_e, m_\mu, m_\tau < 1\,\%$ [K]	G(6)-internal
Neutrino masses	not derived	m_{ν_i} from fixed points [K]	G(6)-internal
ξ connection	Casimir quotient [K]	$\xi = C_2/N_F$ [K]	**closed**

.5 The Actual Connection Question

The question "is ξ the smallest spectral value of V?" is fully answered by the numerical investigation and R84:

[K] ξ is not a spectral value of any G(6) operator – a closed negative finding, not an open bridge. The connection between G(6) and FFGFT lies on a different level:

$$\xi = \frac{C_2(SU(3)_{\text{fund}})}{N_{\text{Fourier}}} = \frac{4/3}{10^4} = \frac{4}{30000} \tag{11}$$

connects the *algebraic* structure of G(6) (Casimir invariant $C_2 = 4/3$, from $N_c = 3$ [B]) with the *geometric* scale of FFGFT (torus mode count $N_F = 10^4$). This is not a G(6)-internal statement, but the bridge formula between both frameworks – R84.

Note on the electroweak group: $SU(2)_L \times U(1)_Y$ is algebraically derived in *FFGFT* (Doc. 321 [B]). Whether Matzke's $G(6, \mathbb{Z}_3\mathbb{C})$ reproduces the same derivation internally is a *G(6)-internal* problem – not an open FFGFT bridge.

.6 ξ from the Casimir Operator of $SU(3)$

The Quadratic Casimir Operator

[B] The quadratic Casimir operator of a Lie algebra $\mathfrak{g}$ is the group invariant

$$C_2(R) = \sum_{a=1}^{\dim \mathfrak{g}} T_a^{(R)} T_a^{(R)}, \tag{12}$$

where $T_a^{(R)}$ are the generators in representation R. For $SU(N_c)$ in the *fundamental* representation:

$$C_2(SU(N_c)_{\text{fund}}) = \frac{N_c^2 - 1}{2N_c}. \tag{13}$$

With $N_c = 3$ (algebraically necessary, Doc. 321, **[B]**):

$$C_2(SU(3)_{\text{fund}}) = \frac{9-1}{6} = \frac{8}{6} = \frac{4}{3}. \tag{14}$$

ξ as the Casimir Value per Fourier Mode

[K] The Fourier mode count of the T^4 torus:

$$N_{\text{Fourier}} = \frac{N_{\text{modes}}}{|\mathbb{Z}_3|} = \frac{30000}{3} = 10^4. \tag{15}$$

The FFGFT geometry parameter follows:

$$\boxed{\xi = \frac{C_2(SU(3)_{\text{fund}})}{N_{\text{Fourier}}} = \frac{4/3}{10^4} = \frac{4}{3 \times 10^4} = \frac{4}{30000}} \tag{16}$$

[B] This formula is exact. Each quantity has a geometric or algebraic justification:

- $C_2(SU(3)_{\text{fund}}) = 4/3$: algebraic from $N_c = 3$, which is algebraically necessary from the $\mathbb{Z}_3$ structure (Doc. 321).
- $|\mathbb{Z}_3| = 3$: order of the orbifold group (Doc. 321).
- $N_{\text{Fourier}} = 10^4$: Fourier mode count of the T^4 torus, from 4-dimensional topology and resolution scale.

Connection to Doc. 009: Two Routes to the Factor 4/3

[K] Doc. 009 shows that the Casimir effect (vacuum energy between plates) delivers the factor $4/3$ from QFT:

$$E_{\text{Casimir}} = -\frac{\pi^2 \hbar c}{720\, a^3} \times \frac{4}{3}. \tag{17}$$

This is the same number as $C_2(SU(3)_{\text{fund}}) = 4/3$ – not a coincidence. The physical Casimir effect (Doc. 009) is the field-theory manifestation of the algebraic Casimir value of the $SU(3)$ algebra: in both cases $4/3$ counts the degree-of-freedom weighting of an $SU(3)$-invariant system.

Table 3: Two routes to the factor 4/3 in ξ

Route	Source	Factor
Casimir effect (physical)	QFT, Doc. 009	$4/3$ from vacuum fluctuations
Casimir operator (algebraic)	$SU(3)$, Doc. 321	$C_2 = (N_c^2 - 1)/(2N_c)$
Result	both	$4/3 \to \xi = 4/30000$

Casimir derivation of ξ [K]

$\xi = C_2(SU(3)_{\text{fund}})/N_{\text{Fourier}}$ derives the numerator 4 in $\xi = 4/30000$ algebraically from the quadratic Casimir operator. $N_c = 3$ is algebraically necessary (Doc. 321 [B]); the factor $4/3$ appears in both the algebraic Casimir and the physical Casimir effect (Doc. 009) – two independent routes to the same number.

.7 Conclusion

Overall status Doc. 324 – $G(6, \mathbb{Z}_3\mathbb{C}) \leftrightarrow$ FFGFT

Verified [B]:

- Trine theorem: $T_k^3 = \mathbf{1}$ in 8-dim spinor representation
- V is rank-1 projector: $\sigma(V) = \{0^{(7)}, 1^{(1)}\}$
- Trine product $(T_1T_2T_3)^3 = \mathbf{1}$ with $\mathbb{Z}_3$ spectrum
- Three sectors, bilinear transitions, confinement as $T_R = 0$
- $C_2(SU(3)_{\text{fund}}) = 4/3$ algebraically from $N_c = 3$

Closed [K]:

- ξ is not a spectral value of V (numerically excluded) [K]
- $\xi = C_2(SU(3)_{\text{fund}})/N_{\text{Fourier}} = (4/3)/10^4 = 4/30000$ algebraically derived – R84 [K]
- The Laplace operator on the vacuum ray gives $\Delta = 6 \cdot \mathbf{1}$ (trivial, since $\Gamma_a^2 = \mathbf{1}$) – no connection to ξ as an eigenvalue; the question is closed as incorrectly posed [K]

Open [S]:

- $SU(2)_L \times U(1)_Y$ in $G(6, \mathbb{Z}_3\mathbb{C})$ not yet derived – **G(6)-internal problem, not an FFGFT problem.** FFGFT has derived $SU(2)_L$ and $U(1)_Y$ from torus geometry (Doc. 321 [B]); the question is only whether Matzke's $G(6, \mathbb{Z}_3\mathbb{C})$ reproduces the same derivation internally.

Remark on the Laplace point: The original question "is there a natural G(6) operator with $\lambda_{\min} = \xi$?" is answered by R84: ξ does not arise from a spectrum within G(6), but from the Casimir quotient C_2/N_{Fourier} – a bridge between algebraic structure (G(6)) and geometric scale (FFGFT torus). These are two different levels; no G(6) operator alone can have ξ as an eigenvalue.

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